# INDEX

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School Name..... Sub. .....

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| | | | | |
| | | | | |

Exp 1 Study of various network command used in Linux and windows

1) arp -a

Show IP Address of your computer alone with IP address and MAC address of your router

O/P:

Interface 192.168.156.1 .... 0x7

| Internet Address | Physical address | type |
|---|---|---|
| 19.2.168.56.255 | f.f.ff.ff.ff. | static |
| 244.00.22 | 01.00.5e002. | static |
| 23.9.255.25.250 | 01.00.5e.ff ff | static |

2) hostname:

Display name of your computer

O/P: System REC

3) ip config/all.:

Display detailed configuration or information about TCP/IP connection and type of ethernet adapter in the system

O/P:

windows IP configuration

Hostname . . System REC

Primary DNS Suffix

Node type

4) nb stat -a:

Display protocol statistic and current TCP/IP connection using NBT

O/P:

NBTSTAT [[-a Remote Name] [-A IP address] [.]

5) netstat:

Display a variety of statistics about a computer active TCP/IP connection

O/P:

Active connection

| Proto | local address | Foreign address | state |
|---|---|---|---|
| TCP | 10.76.9.6.60.49175 | 10.76.96.163.domain | |
| TCP | 10.76.96.60.52234 | 142.250.250.183.http | |
| TCP | 10.76.96.60.58954 | 10.76.96.16.3.chrome | |

6) NSlookup:

Display DNS detail such as IP address of the particular computer

O/P:

Server: unknown

Addres: 10.76.196.163

7) Pathping:

Traces the root to destination address then launches a 25 second test of each routers along the way.

O/P:

Pathping [-ghost-list].[-n m as-in-ng][;-address]

# Linux networking commands

1) ip:

Help from setting up new system and assigning IP to trouble sharing ~~syst~~ existing system

2) ifconfig:

The ifconfig command was a staple in many sysadmin's tool belt for configuration and ~~trou~~ troubleshooting networks

3) mtr:

MTR is a program with CLI that use network-diagnose and trouble shooting tools

4) tcpdump:

The tcpdump command is designed for capling displays packets.

5) Ping:

Ping is a tool that verify. IP. "Connectivities to another tcp/IP computer by sending Internet control message protocol (ICMP) echo request message protocol

(ICMP) echo request message

6) IP config:

flag 73 up lock basic, RUNNING GK

<65556, inet 127.0.0.1 network 255.0.0.0 inet6::1

prefix len 125 scope d x c 10 < list)

7) mtr -no/p:

my traceroute [v. v27]

local host. local domain (.1)

Key - help display mode Restart statistic

| Host | Packet | list |
| --- | --- | --- |
| 1::1 | Rom t<br>0.0 /. | 0.2 |

8) Ping:

Ping 8.8.8.8

64 bytes .8.8.8.8 icmp, seq = 1.tlt. 113

time = 37.4ms

7/10

# Exp-2 Study of different types of Network Cables

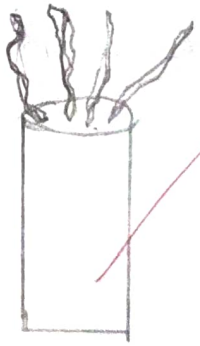
Aim:

To study different types of Network cables. different type of cable used is networking are

1) unshielded Twisted Pair (UTP) cable
2) shielded Twisted pair (STP) cable
3) coaxial cable
4) Fibre optic cable

| Cable Type | Category | maximum Data Transmission | Advantages / Disadvantages | Application/use |
|---|---|---|---|---|
| UTP | Category 3 | 10 bPS | Advantages<br>*) Cheaper in cost<br>*) Easy to install as they have a smaller overall diameter<br>· Disadvantages<br>*) more prone to (EMI) Electromagnetic interference and noise | 10 Base-T Ethernet |
| | Category 5 | upto 100 mbPS | | Fast Ethernet, Gigabit Ethernet |
| | category 5e | 1GbPS | | Fast Ethernet, Gigabit Ethernet |
| STP | category 6,6a | 10GbPS | Advantages<br>*) Shielded<br>*) Faster than UTP.<br>*) Less Susceptible to noise and interference<br>Disadvantages<br>*) Expensive<br>*) Greater installation effort | Gigabit Ethernet, 10G Ethernet (55 m) widely used in data centres |
| SSTP | category 7 | 10 GbPS | | Gigabit Ethernet 10G Ethernet (100 m) |

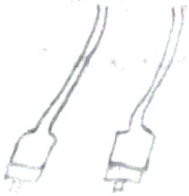
| | | | | |
|---|---|---|---|---|
| Coaxial Cable | RG-6<br>RG-59<br>RG-11 | 10-100 mbps | *) High bandwidth<br>*) Immune to interference<br>*) Low Loss bandwidth<br>*) Versatile<br>Disadvantages<br>*) Limited distance<br>*) cost<br>*) Size is bulky | Speed of signal is 500m Television network High speed internet connections. |
| fibre optics cable | Single mode<br>multi mode | 100 Gbps | Advantages:<br>*) High speed<br>*) High bandwidth<br>*) High security<br>*) Long distance<br>Disadvantages<br>*) Expensive<br>*) Requires skilled installers | maximum distance of fibre optics cable is around 100 meters |









| | |
|---|---|
| Coaxial cable |  |
| Fibre optic cable |  |

Result:

Thus the study of different types of Networking cables is done.

8/10

# Exp: 3 Cisco Packet Tracer

Aim:

To study the Packet tracer tool installation and user interface overview and understood the packet tracer environment to design a simple network

Procedure:

A) Network using hub:

i) open cisco packet tracer

ii) from the network component box add :

* 4 PCs
* 1 hub

iii) ~~Sele~~ Soket connection

iv) select copper straight through cable

v) connect all 4 pcs to all the hub

vi) configure the IP address and Subnet mask of the PCs

PC → desktop → IP configuration.

vii) use the PDV / message icon to send a packet from the PC to another.

viii) observe the packet flow in the simulation

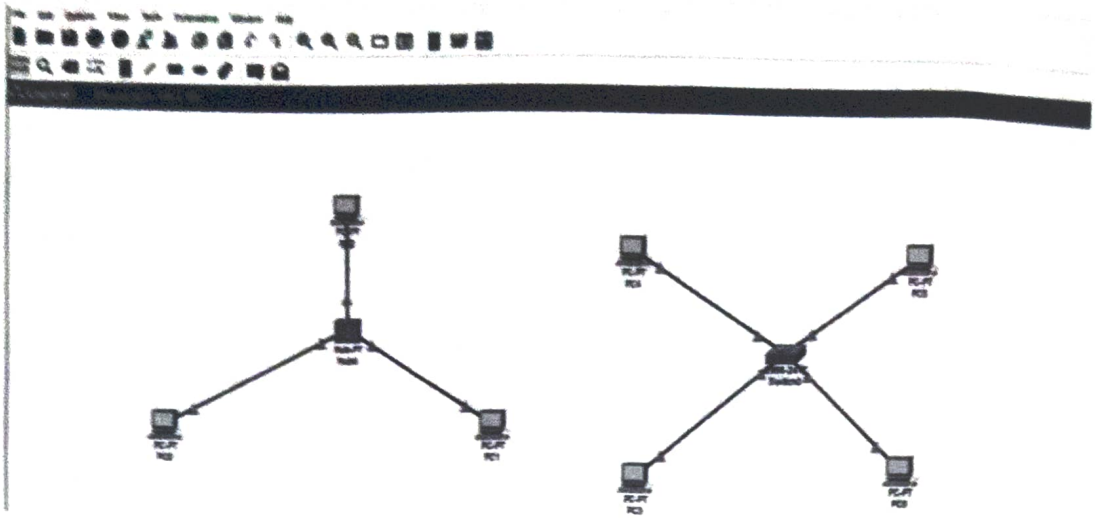
Example of IP configuration:

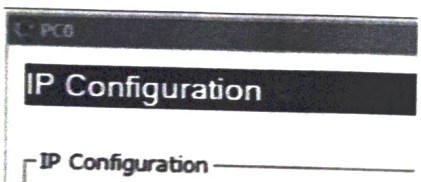
| device | IP address | Subnet mask |
|---|---|---|
| $Pc_0$ | 192.168.1.1 | 255.255.255.0 |
| $Pc_1$ | 192.168.1.2 | 255.255.255.0 |
| $Pc_2$ | 192.168.1.3 | 255.255.255.0 |
| $Pc_3$ | 192.168.1.4 | 255.255.255.0 |

Part - B - networks using switch

i) add 4PCs and 1 switch
ii) connect each Pc to the switch using copper straight through cable
iii) configure IP address and subnet mask
iv) sent a PDV from 1 pc to another
v) observe the packet flow using simulation mode.

| hub | Switch |
|---|---|
| broadcast the incoming frame to connect ports | forward the frame towards the distribution port after mac address |
| does not make forwarding decision based on mac address | uses a mac address table for forwarding |
| more unnecessary traffic | more efficient communication |
| collision domain is shared | each switch port provide a separate collision domain |






Result:

Thus the simple networks using hubs and switches where designed and simulated successfully.

exp:4 LAN using Switch

Aim:

to setup and configure LAN using a switch and ethernet cables in the laboratory.

procedure:

1) plan the LAN topology according to the network requirement

2) take the following devices

*) 4PCS
*) 1Network switch
*) 4 ethernet cables

3) connect PC1, PC2, PC3, PC4 to the switch using ethernet cable

4) configure the IP address of each PC:

PC1 -> 10.1.1.1
PC2 -> 10.1.1.2
PC3 -> 10.1.1.3
PC4 -> 10.1.1.4

5) to configure each PC in packet tracer

PC -> desktop -> IP configuration -> static

6) enter the corresponding IP address and Subnet mask.

7) Configure the network Switch with IP address 10.1.15 and subnet mask 255.0.0.0

8) open command prompt on each PC

9) use the ping command to check connectivity between PCs and switch

10) verify that the ping is successful

11) select a folder on one computer enabling sharing for computer on the same LAN

12) from another computer access the shared folder to verify file sharing

8/10

Result :

Thus, the LAN was successfully configured using switch and ethernet cable

exp:5 experiment on packet capture tool: wireshark

Aim:

To use packet capture tool capturing and analysing packet using this tool.

Packet sniffer:

*) Sniffes messages sent or receive by the computer
*) store and display the content of various protocol
*) work as a passive program
*) does not sent packet by itself.

wireshark:

it captures network and allow the user to the protocol and fields within those packets

*) capture network traffic
*) decode packet protocols
*) define capture and display packet
*) analyze n/w problems

procedure:

step 1:

start wireshark

step 2:

Select network interface

=) select the required network interface, such as active wi-fi or ethernet interface

Step 3 : Observe captured packets

=> The captured packets are displayed in the packet list pane. each line represents one packet captured from the network

Wireshark Packets :

1) packet list pane :

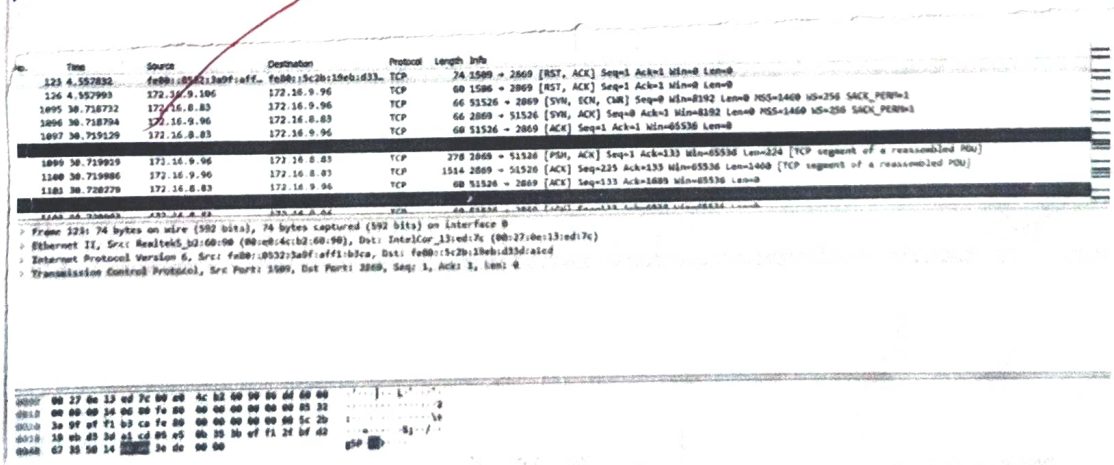
=> The packet list pane displayed all packets in the current capture

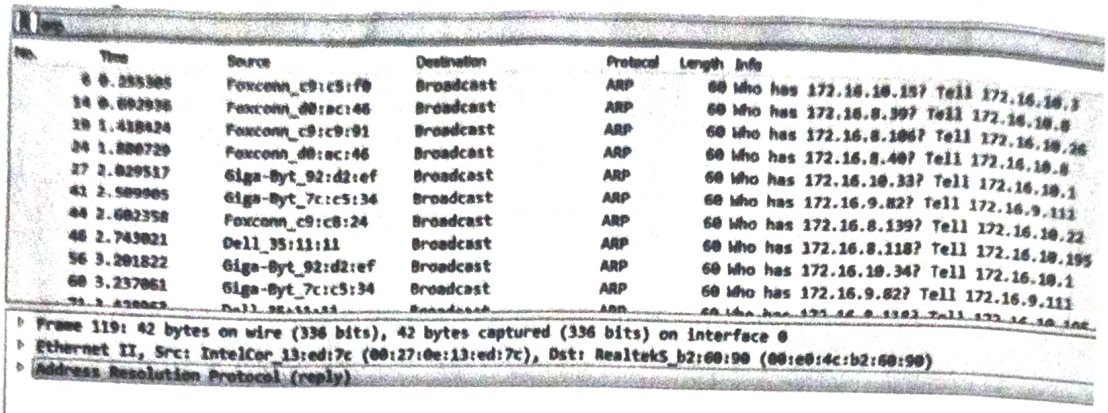
2) packet details pane :

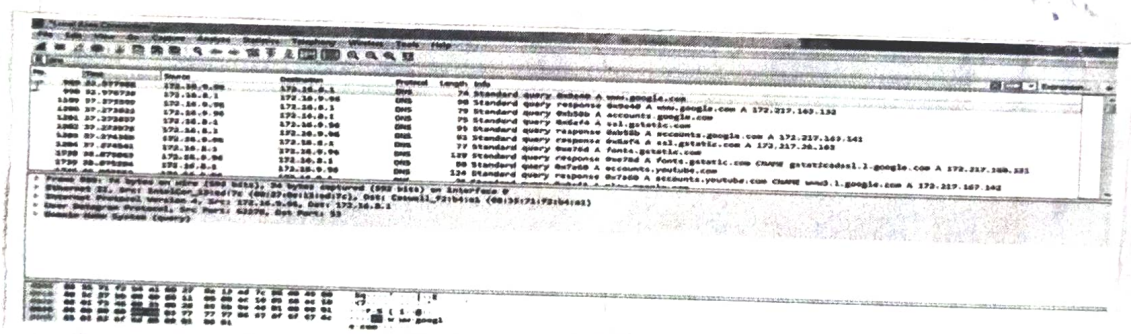
The packet details pane displayed detailed information about the selected packet

3) packet bytes pane :

it displayes the actual data of the selected packet in hexdump format.





Result :

This the above experiment has been executed Successfully.

exp: 6:

error correction at data link layer (hamming code)

Aim:

To write a program to implement error detection and correction using hamming code. and verify error correction.

Sender program:

1) get a text message from the user
2) convert the text into binary
3) apply the hamming code to the binary data
4) add the required redundant bits
5) store the routing data in the file called channel.

code:

```
def hamming_code(data):
    encode = ''
    for i in range(0, len(data), 4):
        d = data[i : i+4]
        if len(d) < 4:
            d = d.ljust(4, '0')
        d1, d2, d3, d4 = map(int, d)
        p1 = d1 ^ d2 ^ d4
        p2 = d1 ^ d3 ^ d4
        code = str(p1) + str(p2) + str(d1) + str(p4) +
               str(d2) + str(d3) + str(d4)
```

```
encode += code
return encoded
text = input ["enter text : "]
binary = "".join (format (ord(c), "08b") for c in
                  text)
print ("binary data", binary)
encoded_data = hamming_encode (binary)
with open ("channel.txt", "w") as file:
print ("hamming encoded data :", encoded_data)
print ("hamming encoded data :", encoded_data)
print ("data saved to channel.txt")
```

receiver program:

=> reads data from text
=> check every 7 list hamming block
=> find the position using syndrome
=> corrects the error remove parity bit

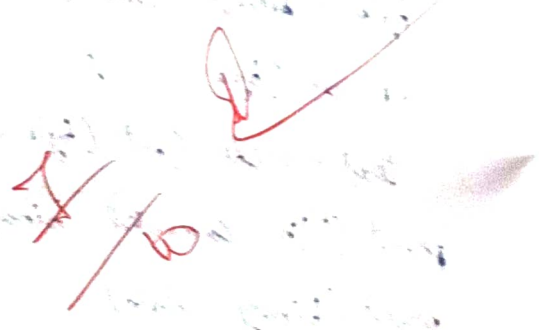
Code:

```
def hamming_decode (data):
    corrected
    error_found = false
for i in range (0, len (data), 7)
block = list (map (int, data [i:i+7]))
if len (block) < 7
continue
p1, p2, d1, p4, d2, d3, d4 = block
```

```
d1 = p1 ^ d1 ^ d2 ^ d4
d2 = p2 ^ d1 ^ d3 ^ d4
d4 = p4 ^ d2 ^ d3 ^ d4
error - position = d1 + (d2 * 2) + (d4 * 4)

if error position != 0
    error - found = time
    actual - position = i + error - position
    print ("error detected")
    block [error - position - 1] ^ = 1
    corrected += str (block [2])
with open ("channel.txt", "r") as file.
    channel - data = file. read CS. strip ()
    print ("data received")
    print (channel data)
for i in range (0, len (binary - data), 8)
    byte = binary data (i: i+8)^n
    if len (byte) == 8.
    text + = chr (int (byte, 2))]
print ("\n corrected binary data:")
print (binary - data)
if error - fecontend
        print ("error detected")
else
    print ("\n no errors")

input:
    enter text HELLO
```



Result :

Thus the above exercise has been executed successfully

exp-7

# Sliding window protocol

Aim

writing a program to implement flow control at data link layer using sliding window protocol.

procedure:

sender program

1) input window size from the user
2) input a text msg frame the user
3) consider character per frame
4) create a from with following fields
5) send the frames
6) wait for the acknowledgement from receiver
7) reader file called receiver - buffer
8) check ack field for the acknowledgement number

receiver file:

1) reader field called sender - buffer
2) check the frame no.
3) if the frame no. are as expected write the appropriate code in the receiver - buffer file

C - program:

```
# include <stdio.h>
int main ( )
{
  int n, window, i;
  printf ("enter no. of frames");
  scanf ("%d", &n);
  printf ("enter window size");
  Scanf ("%d" & window);
  printf ("\n sliding window transmission \n");
```

```
for (i=1; i<=n; i++)
  {
   if (i % window == 0 || i == n)
   {
   printf ("Ack received for frame %d", i);
   printf ("\n");
   }
  }
printf ("all frame transmitted successfully")
return 0;
}
```



result:
Thus the above curve has been sketched successfully

ex:8 NMAP live host discovery

Aim:

To discover live host using NMAP such as ARP, ICMP, TCP and UDP on the try hackme NMAP room.

1) ARP - icon:

ARP used to discover hosts on the same subnet

NMAP can perform an ARP host discovery using

NMAP - PR - Sn target

2) ICMP icon:

it can be used to check whether a host is acti

An ICMP echo request is sent to the target if the host is active it normally responds with an ICMP Echo replay.

eg:

nmap - PE - Sn TARGET

3) TCP SCAN:

can also use TCP packets to discover live host

A TCP ping node npm packet to a specified TCP port A response is dictates that the host is reachable

eg:

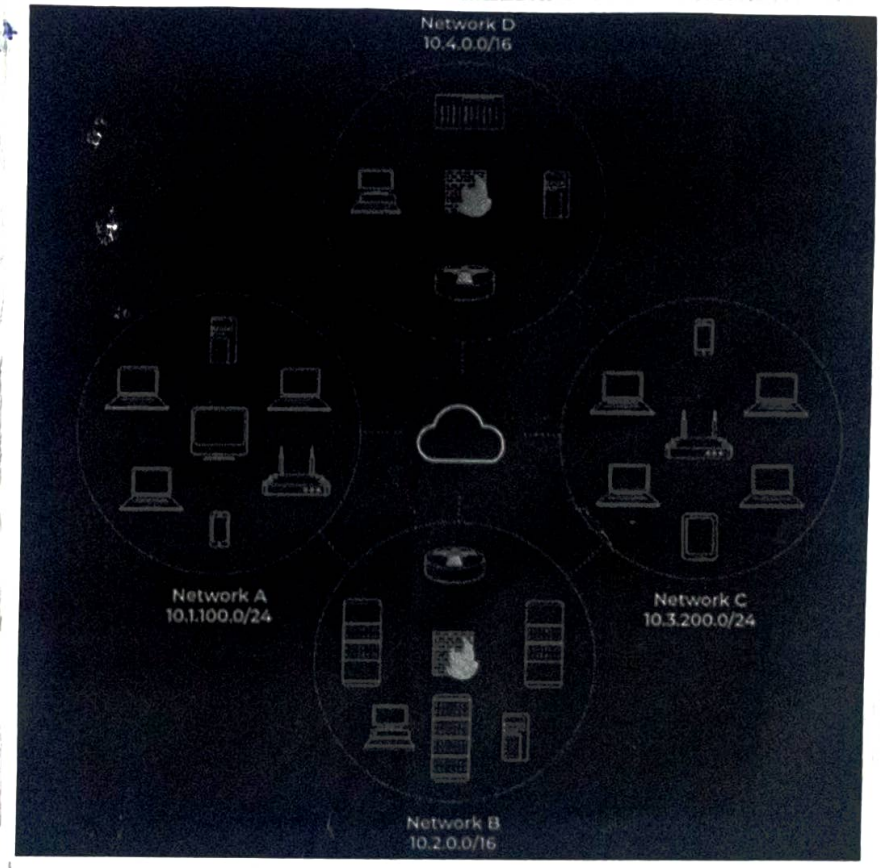
nmap - PS 23 target

4) UDP icon:

NMAP can use UDP packets for host discovery when a UDP packet is sent to a closed port, the target may respond with an ICMP that the port unreachable message. This response can indicate that the target system is active.

procedure :

1) open kaylinux nmap room
2) start the required machine / attackbox
3) identify the target ip address
4) use nmap for host discovery
5) perform ARP, ICMP, TCP or UDP discovery as request
6) observe the host response
7) record the discovered live host
8) verify its results



Send Packet

From:

computer1

To:

computer1

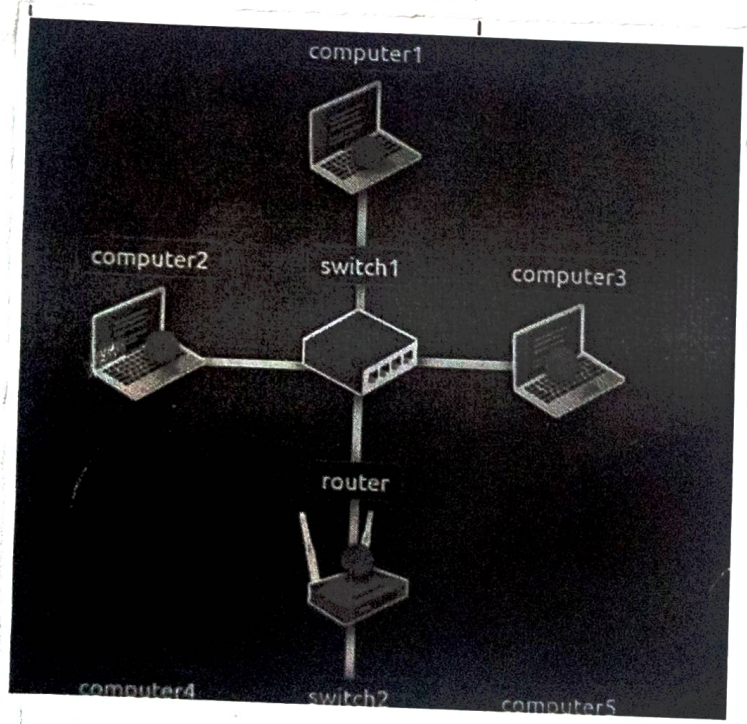
Packet Type:

arp_request

Data:

computer6

Send Packet




8/10

Results:

Thus the above exercise has been executed successfully.

# Experiment - 9

Aim:

Implementation of subnetting in CISCO packet tracer simulator

classless IP subnetting is a technique that allows for more efficient use of IP address by allowing for subnet mask

create a network topology.

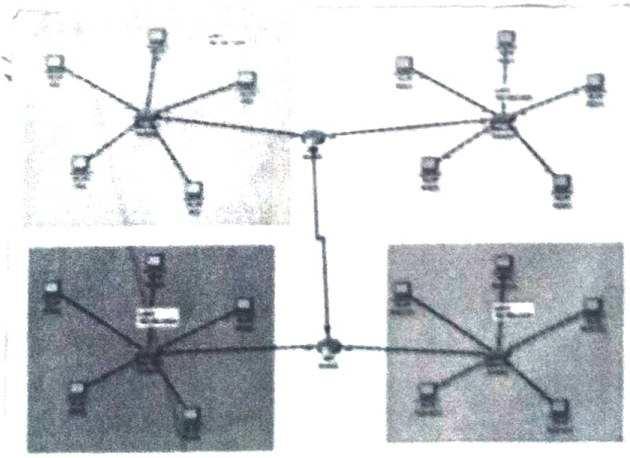
To create a network topology in packet tracer, select "New" button in top, then select network & generic

Adding the device:

To add a device, select the device from bottom left and drag it onto network topology

Subnetting:

To subnet the network address of 192.168.1.0/20 to provide enough space for atleast 5 address for end devices, switch and router



configuring the devices:

Right click on router and select "CLI". This will open by configuring the router. Right click CLI enter following commands

# enable
# interface FastEthernet 0/0
# ip address [IP address] [subnet mask]
# no shutdown
# exit
interface Fast ethernet 0/1
# ip address [IP address] [subnet mask]
no shutdown
exit.

Result:

The implementation has been completed successfully.

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Exp - 10(a)

Aim:

Internet working with routers in Cisco PACKET Tracer simulator

procedure:

Step 1: configure router 1.

Step 2: configuring PC's

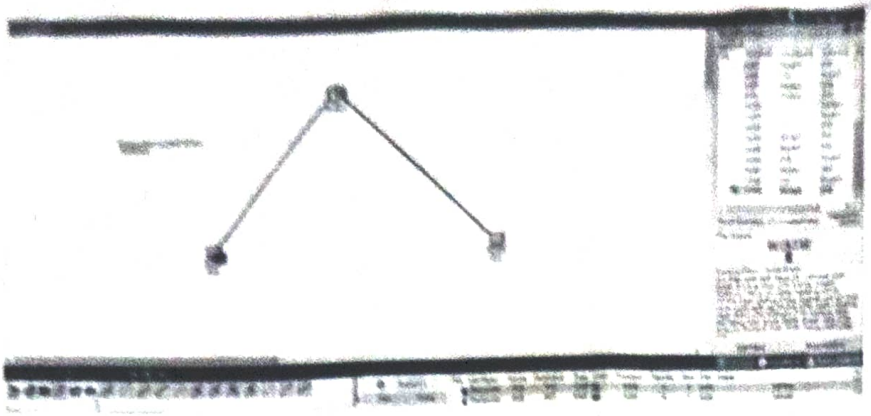
step 3: connecting PC with router

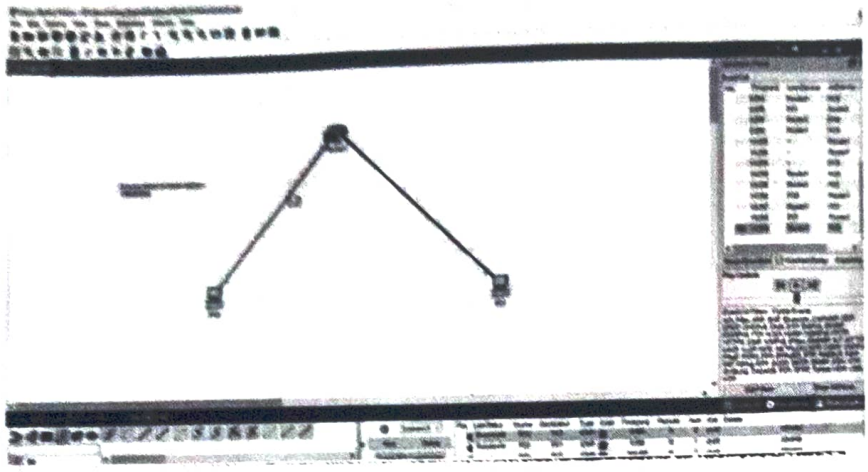
router configuration table.

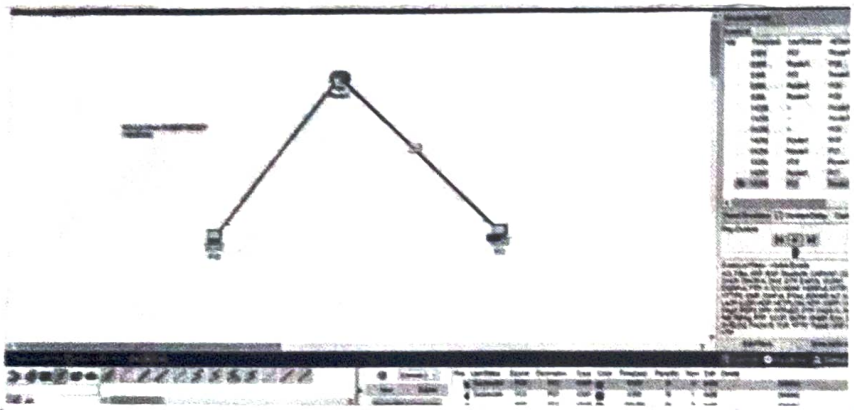
| Device name | IP address ethernet 0/0 | Subnet mask | IP address ethernet 0/1 | Subnet mask |
|---|---|---|---|---|
| Router | 192.168.10.1 | 255.255.255.0 | 192.168.20.1 | 255.255.255.0 |

PC configuration table:

| Device name | IP address | Subnet mask | Gateway |
|---|---|---|---|
| PC0 | 192.168.10.2 | 255.255.255.0 | 192.168.10.1 |
| PC1 | 192.168.20.2 | 255.255.255.0 | 192.168.20.1 |







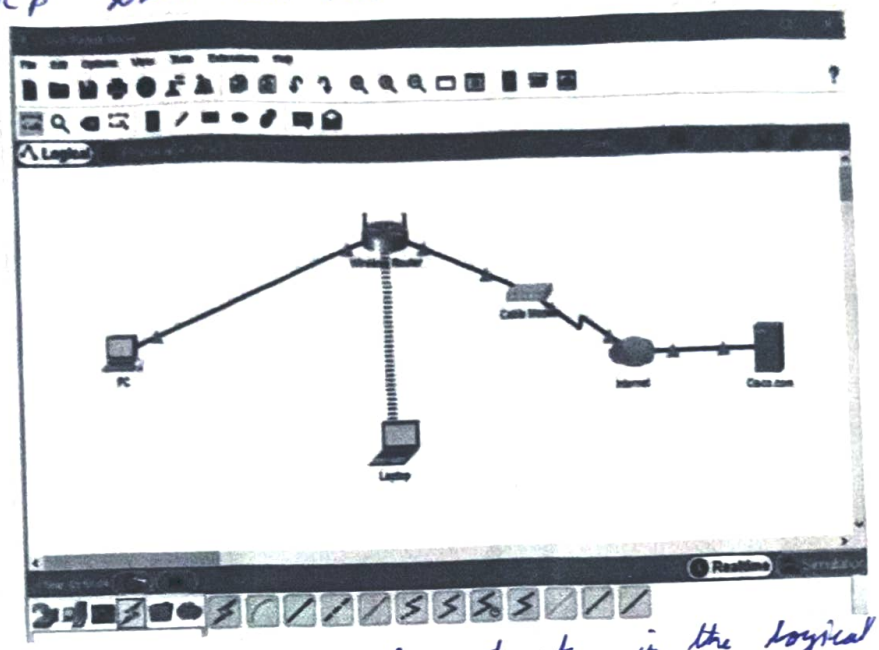
Result :

- The interactworking units routes has been completed succ

Exp: 10 (b)

Aim:

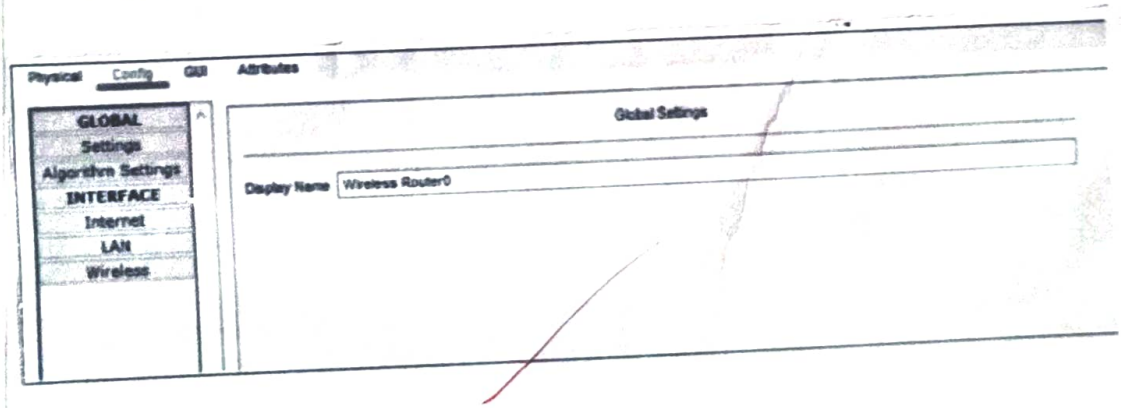
Design and configure an internetworking using window router, DHCP server and internet cloud.




part 1: Build a simple network in the logical topology workspace

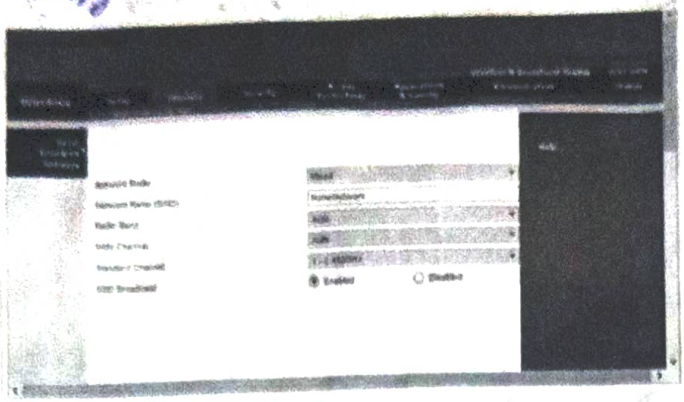
→ step 1: launch packet tracer.

→ step 2: Build the topology.

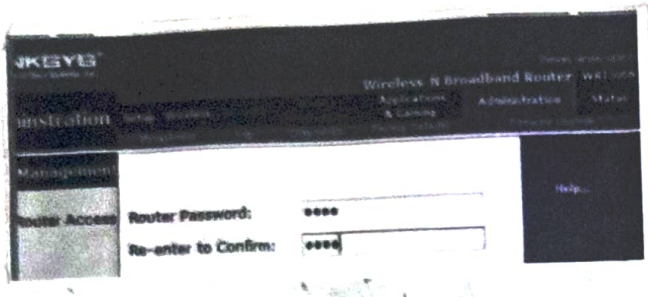


part 2: configure the network device

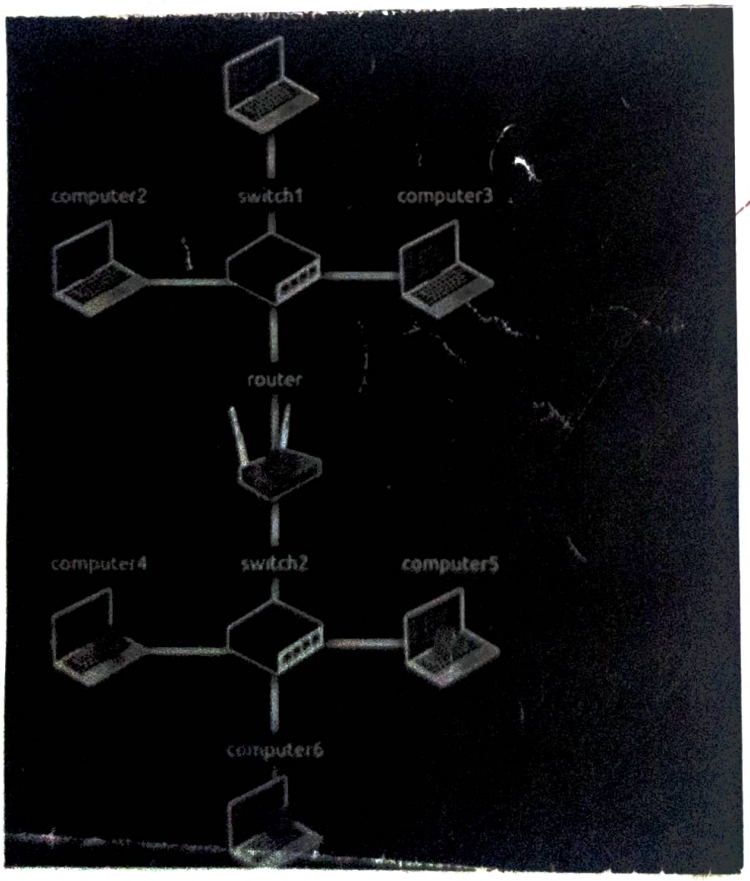
⇒ step 1: configure the wireless router.



step 2:




step 3: configure the laptop



step 4 configure the PC



step 5 configure interface

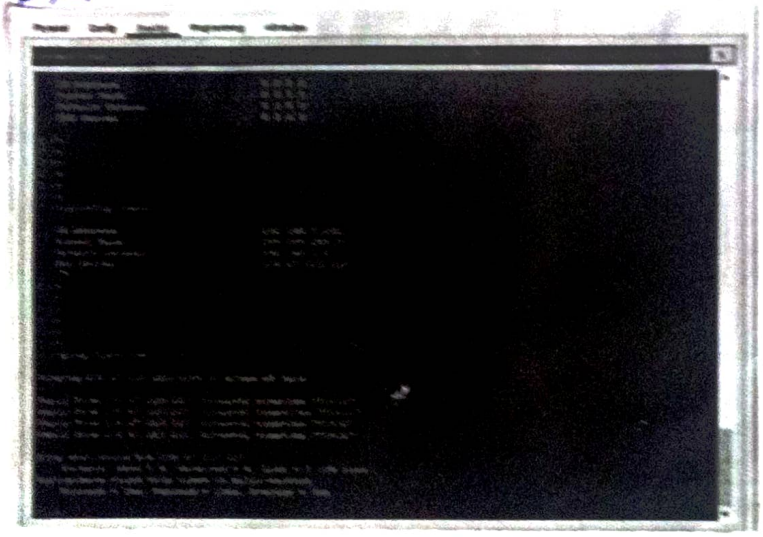
step 6: Configure the Cisco.com server



part-3: verify connectivity

step 1: Refresh the IPV6 setting on PC

a) verify that PC is receiving IPV6 configure information from DHCP



b) Test connectivity to Cisco.com server from PC

Result:

This experiment has been completed successfully

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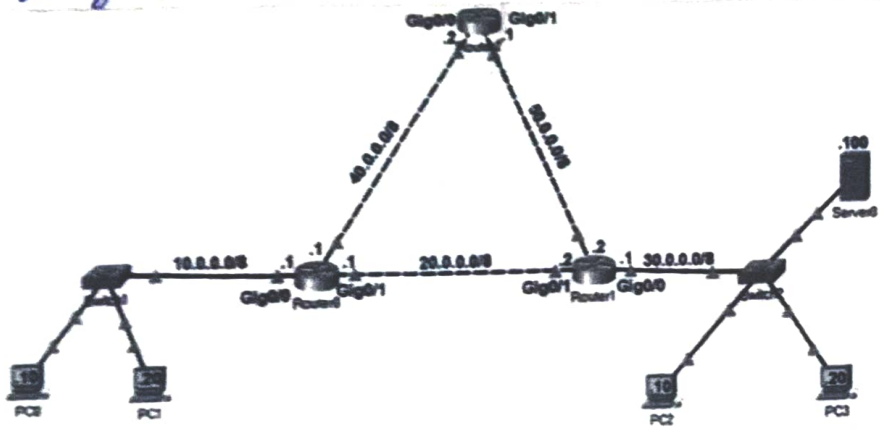
Exp - 11(a)

Aim:

Simulate static routing configuration using CISCO packet tracer

procedure:

Setting up a practise lab




Creating, adding, verifying Italk routes

connected network of each router

| Router | Available on local | Available on other router |
|---|---|---|
| Router 0 | 10.0.0.0/8<br>20.0.0.0/8 | 30.0.0.0/8,<br>50.0.0.0/8 |
| Router 1 | 20.0.0.0/8,<br>30.0.00/8 | 10.0.0.0/8,<br>40.0.0.0/8 |

Router 0 configuration.

Router 0 enable

Router # configure terminal

Router (config) # ip route 30.0.0.0 255.0.0.0 20.0.0.2

Router(config) # ip route 30.0.0.0 255.0.0.0 40.0.0.2

Router (config) # exit

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2 10 Primary route
Router(config)#ip route 30.0.0.0 255.0.0.0 40.0.0.2 20 Backup route
Router(config)#ip route 30.0.0.100 255.255.255.255 40.0.0.2 10 Primary route
Router(config)#ip route 30.0.0.100 255.255.255.255 20.0.0.2 20 Backup route
Router(config)#ip route 50.0.0.0 255.0.0.0 40.0.0.2 10 Primary route
Router(config)#ip route 50.0.0.0 255.0.0.0 20.0.0.2 20 Backup route
Router(config)#exit
Router#show ip route static
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [10/0] via 20.0.0.2
S       30.0.0.100/32 [10/0] via 40.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2

Router#
```

Router adds only primary routes to the routing table.

Router 1 configuration

Router enable

Router # configure terminal

Router (config) # ip route 10.0.0.0 255.0.0.0 20.0.0.10

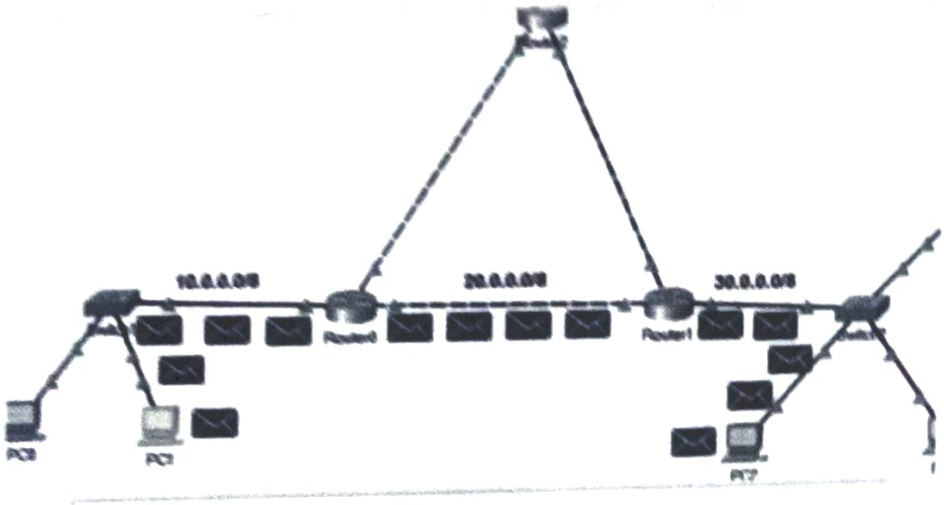
Router (config) # ip route 10.0.0.0 255.0.0.0 50.0.0.1.20

Router (config) # exit

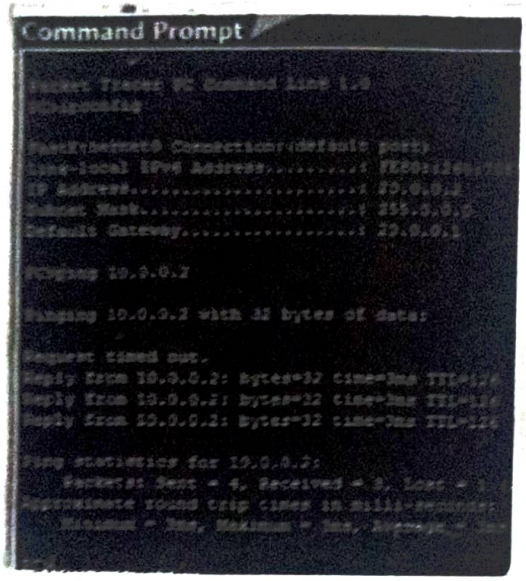
```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with
Router(config)#ip route 10.0.0.0 255.0.0.0 40.0.0.1
Router(config)#ip route 30.0.0.0 255.0.0.0 50.0.0.2
Router(config)#exit
Router#show ip route static
S     10.0.0.0/8 [1/0] via 40.0.0.1
S     30.0.0.0/8 [1/0] via 50.0.0.2

Router#
```

verifying static routing



To simulate the failure of main route you can delete the link between router 0 and router 1




Result:

This experiment has been completed successfully.

Experiment - 11 (b)

Aim:

Simulate RSP using Cisco packet tracer.

Initial IP configuration:

| Device | Interface | IP config | Connected with |
|---|---|---|---|
| PC 0 | Fast ethernet | 10.0.0.2/8 | Router 0's Fa0/1 |
| Router 0 | Fa 0/1 | 10.0.0.1/8 | PC0's Fast ethernet |
| Router 0 | S0/0/1 | 192.168.1.254/30 | Router 2's S0/0/1 |
| Router 1 | S0/0/1 | 192.168.1.259/30 | Router 0's S0/0/0 |
| Router 1 | S0/0/1 | 192.168.1.256/30 | Router 0's S0/0/1 |
| PC 1 | Fast ethernet | 20.0.0.2/30 | Router 2's Fa0/1 |

Assign IP address to PCs

Double click PC's and click desktop menu item and click IP config.

Router 0:

Router (config) # interface fast ethernet 0/0
Router (config-if) # ip address 10.0.0.1
255.0.0.0 Router (config) # no shutdown
Router (config-if) # exit

Router

Router 1.

Router > enable
Router # configure
Router (config) # interface serial 0/0/0
Router (config-if) # no shutdown
Router (config-if) # exit

Router 2

Router > enable
Router # configure
terminal
Router (config) # interface fast ethernet 0/0
Router (config-if) # ip address 20.0.0.1
255.0.0.0 Router (config-if) # no shutdown
Router (config-if) # exit

Access the command prompt of Pc, and use ping command to test connectivity from Pc0

Result:
The experiment has been completed successfully.

experiment 12(a)

Aim :

Implement echo client server using TCP / UDP sockets

Algorithm :

TCP server

1) start the program
2) create server socket using a port number.
3) wait using accept ()
4) create I/p & O/p streams.
5) Receive message from client
6) send same back to client
7) stop

TCP client

1) start the program
2) create a socket & connect to server using port no.
3) create I/p . o/p stream
4) Read a message from user
5) send same to server
6) Receive the echoed message
7) stop

I/P :

Enter message : Hell · server

O/P:

server started
waiting for client
client connected
message from client: Hello server.

Result:

The program has been executed successfully.

Experiment - 12 (b)

Aim:

To implement a chat - client server application using TCP/UDP socket

Algorithms:

TCP chat server

1) Start the program
2) Create server socket with
3) wait using accept ( ).
4) Create I/p & O/p streams
5) Receive message from client
6) Display message
7) send reply to client
8) close & socket & stop program

TCP chat client

1) start the program.
2) Create socket & connect to server
3) Create I/p & O/p streams
4) Read a message from client server
5) Sent message from client server
6) Display reply
7) Close the socket and stop program

I/P:

Client : Hello

Server : Hi

Client : How are you?

server : I am fine

Client : Exit

O/P:

Server started

waiting for client

Client connected

Result:

The program has been executed successfully.

Experiment - 13

Aim:

Implement your own ping program

program:

```
# include <stdio.h>
# include <string.h>
# include <stdlib.h>
int main () {
    int sock;
    Char buffer[10,24];
    struct sock add in addr;
    struct icmphdr icmp;
    Sock = socket (AF_INET, SOCK_RAW, IPPROTO_ICMP);
    if (sock < 0){
        perror ("Socket");
        return 1;
    }
    memset (& icmp, 0, size of (icmp));
    icmp.type = ICMP_echo;
    icmp.code = 0;
    sendto (sock, & ICMP, size of (icmp), 0, struct sockaddr *)
    & addr, size of laddr));
    printf ("Reply received from 8.8.8.8\n");
```

Result:

The program has been executed successfully.

Experiment - 14

Aim:

Write a code using RAW sockets to implement packet sniffing.

program:

```
#include <stdio.h>
#include <unistd.h>
#include <arpa/inet.h>
int main () {
    int sock,
    char buffer [65536];
    Struct sockaddr saddr;
    int len = size of (saddr);
    while (1) {
    int n = recvfrom (sock, buffer, size of (buffer), 0, addr,
    (socklen_t *) & len);
    if (n > 0) {
        struct iphdr * ip = (struct iphdr *) buffer,
    printf ("packet received \n");
    }
    close (sock);
    return 0;
    }
```

Result:

The program has been executed successfully.

Experiment - 15

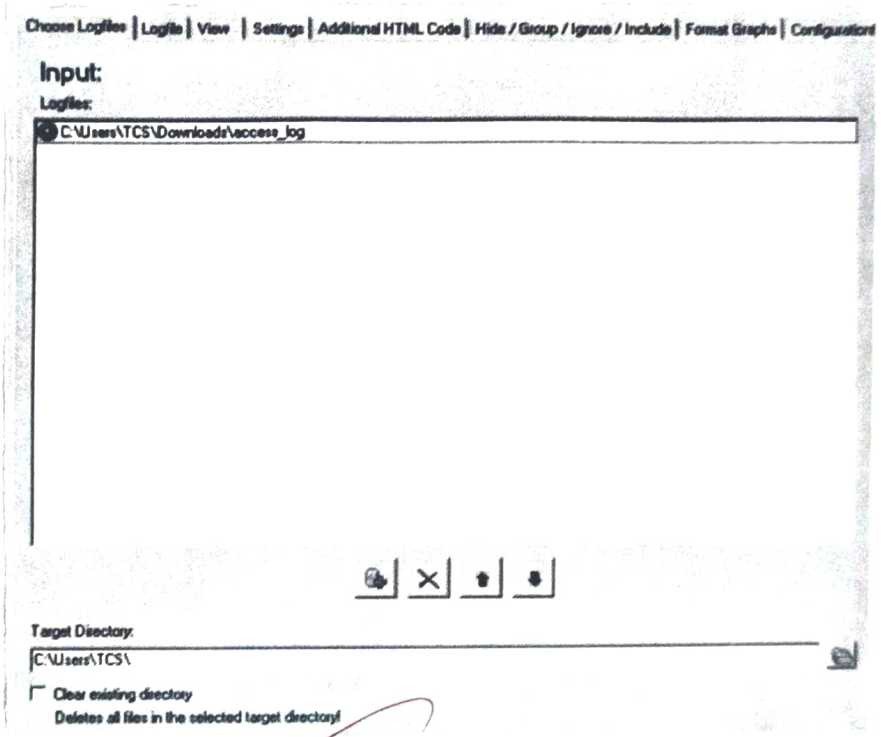
Aim : To analyze the different type of web logs using webalizer tool

procedure :

step 1 : Run webalizer window version

step 2 : Input web log file

step 3 : press run webalizer.




10/10